

# NADA MOHAMED

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## Summary

Computer Science graduate (2026) specializing in Artificial Intelligence and Machine Learning, with an engineering background spanning intelligent agents, LLM fine-tuning, computer vision, full-stack development, and blockchain. Have shipped 15+ projects end to end and taught programming to 30+ students — equally at home building systems and explaining them. Comfortable across the stack in Python, Go, Java, and C/C++.

## Education

**Arab Academy for Science, Technology & Maritime Transport** **Sep 2022 – Jun 2026**

*B.Sc. Computer Science (Dual Degree), Graduated with Honours – GPA: 3.69 / 4.00* *Cairo, Egypt*

- Dual-degree program with the University of Northampton (UK). Coursework: Machine Learning, Computer Vision, NLP, Algorithms & Optimization, Parallel Computing, Compilers, Databases, Robotics.

## Experience

**Dell Technologies** **Aug 2025 – Sept 2025**

*Summer Intern* *Cairo, Egypt*

- Led the Academy Project from concept to business proposal, applying design thinking to Dell's onboarding.
- Completed workshops in front-end development, containerization (Docker, Kubernetes), and software engineering practice.

**DEPI – Microsoft ML Engineering Curriculum** **Nov 2024 – May 2025**

*AI/ML Engineer Intern* *Zamalek, Egypt*

- Completed a 7-month intensive AI/ML program covering NLP, generative AI, computer vision, and deployment.
- Delivered milestone projects and a capstone integrating multimodal workflows and agent orchestration.

**Megz** **Aug 2020 – Nov 2021**

*Computer Science Instructor* *Cairo, Egypt*

- Taught programming fundamentals to 30+ students, improving proficiency in Python and Java.
- Supported the lead teacher in classroom management, grading, and student progress tracking.
- Mentored students in debugging and problem-solving; personalized guidance improved small-group performance 20%.

## Selected Projects

**Payramid — Blockchain Payroll for Egypt** 🌐🔒 **Sept 2025 – Present**

*Go, TypeScript, Blockchain, Docker · Graduation Project*

- Payroll orchestration and immutable audit layer letting employers disburse salaries over Egypt's national payment rails with verifiable proofs of payment, within local banking and data-protection regulation.
- Authored the thesis and the IEEE 830 SRS start to finish; currently revising the accompanying research paper. Platform source is private — links go to the public landing page and its repository.

**Music Genre Detection** 🌐 **Dec 2025**

*Python, PyTorch, Librosa, Flask, Captum*

- CNN-LSTM attention model classifying audio into 10 genres from mel-spectrograms, served through Flask.
- Grad-CAM and Integrated Gradients explainability, a pytest suite with coverage, and Sphinx API docs.

**Pixel Paradise** 🌐 **May 2024 – Feb 2025**

*JavaScript, PHP, HTML/CSS, Python*

- Retro browser arcade built with a small team over 126 commits: playable games, user accounts and sessions, and score tracking behind a handcrafted pixel-art interface. Core developer across gameplay and site features.

## Technical Skills

**Languages:** Python, Go, Java, C/C++, JavaScript, TypeScript, PHP, Dart, R, Bash, SQL, HTML/CSS

**AI / ML & Data:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Librosa, HuggingFace Transformers, Whisper; NLP, Generative AI & LLM fine-tuning, Computer Vision, Reinforcement Learning, Explainable AI (SHAP, Grad-CAM)

**Web & Application:** Flask, FastAPI, React, Tailwind CSS, Vite, JavaFX, Tkinter, Flutter, SwiftUI, PyWebView

**Systems, Cloud & DevOps:** Docker, Kubernetes, Git, GitHub Actions (CI/CD), MySQL, AWS RDS, SQLite; parallel computing, compilers, algorithm design, Arduino

**Blockchain & Security:** Hyperledger Fabric, Chaincode (Go), X.509 PKI, cryptography, threshold (MPC) custody

**Soft Skills:** Mentorship & instruction, team collaboration, technical writing & documentation, analytical thinking, communication, ownership

## Certifications, Awards & Languages

🏆 **First Place, AI/ML Project Submission – AASTMT** • **Microsoft ML Engineer Course – DEPI**

🏆 **AI & Machine Learning Course – Universitat Autònoma de Barcelona** • **CS Camp – Carnegie Mellon University**

**Languages:** Arabic (Native) • English (Fluent) • Spanish (Intermediate)